

Probing optical spin with circularly polarized Raman scattering in silicon metasurfaces

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Nanophotonic metasurfaces are versatile platforms to manipulate light–matter interactions at the nanoscale, with polarization emerging as a particularly powerful degree of freedom [1]. In high-refractive-index dielectric nanostructures, Mie resonances enable strong field confinement and enhancement without the losses associated with plasmonics, making them attractive platforms for spectroscopy and sensing. While polarization control in the far field has been extensively explored, the polarization properties of optical near fields remain difficult to access experimentally. This limitation is particularly relevant for applications such as chiral sensing, where the interaction strength is governed by the local polarization state rather than the propagating field [2].

Raman scattering provides a promising route to probe near-field polarization in dielectric nanostructures. In bulk crystalline materials such as silicon, the polarization of Raman scattering is well described by the Raman tensor and exhibits a predictable response under circularly polarized excitation [3]. However, in nanophotonic resonators the internal field distributions become highly non-uniform and resonance-dependent, potentially altering both the efficiency and polarization state of the Raman emission. Despite extensive work on Raman enhancement in dielectric metasurfaces, the polarization properties of resonantly enhanced Raman scattering—and their connection to the underlying near-field structure—have remained largely unexplored [4].

In this work, we theoretically and experimentally investigate circularly polarized Raman scattering from silicon-on-sapphire disk metasurfaces under circularly polarized excitation in a backscattering geometry. By systematically varying disk diameter, height, and excitation wavelength, we measure both the Raman enhancement factor and the degree of circular polarization (DOCP) of the scattered Raman signal. While certain metasurfaces preserve a relatively high DOCP comparable to bulk silicon, others exhibit a pronounced depolarization despite strong Raman enhancement. Through numerical simulations and multipole decomposition, we identify magnetic and electric quadrupole resonances as the primary origin of this polarization loss. These modes produce complex near-field distributions with spatially varying optical spin, leading to an incoherent mixture of polarization contributions in the Raman emission. Our results establish polarization-resolved Raman scattering as a sensitive and experimentally accessible probe of near-field polarization and optical spin in dielectric nanophotonic systems, with important implications for chiral nanophotonics.

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